GlandersWatch: A Modular Platform for Glanders Risk Prediction, Retrieval-Augmented Explanation and Owner Alerting

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Abstract

Glanders is a bacterial zoonosis of equids caused by *Burkholderia mallei*. It is notifiable, frequently fatal, and remains endemic in parts of Africa, Asia and South America. Indian surveillance between 2019 and 2023 identified 728 seropositive equids across 138 districts, with strong geographic concentration. A district officer and an equine owner both need more than a retrospective case count: they need a usable forecast for the next period, a way to ask what official guidance requires, and messages sparse enough that people still read them. The published machine learning literature on this disease is confined to laboratory identification of the organism; no population-level geographic risk model, and no owner-facing warning path, has been reported.

This report presents GlandersWatch as three independently deployable components. A calibrated risk engine forecasts whether glanders is documented in a country in year $t{+}1$ from features observed in year t. A retrieval-augmented generation (RAG) layer answers questions from officers or owners using passages retrieved from official guidance: every claim must cite a retrieved passage, and the layer refuses when the corpus does not support an answer. An alert service decides whether a message should be sent at all (thresholding, hysteresis, suppression and deduplication) so volume stays low enough that messages are still read. Components talk through versioned schemas rather than shared internals; none hardcodes the disease; and each can degrade without taking the others down.

The risk engine combines four public datasets not previously applied to glanders into a country-year panel: a text-mined occurrence corpus from Europe PMC, FAOSTAT livestock stocks, World Bank development indicators and NASA POWER reanalysis climatology. 6,700 country-year rows reduce to 2,880 modelling observations across 160 equine-holding countries from 2005 to 2023, with 169 positive labels. Eight of 42 numeric candidates are retained by mean absolute SHAP on the training years alone, and six model families are compared under temporal and leave-country-out validation. On unseen years the strongest precision-recall AUC is 0.495 against a 5.87% base rate; on unseen countries the ranking inverts and Logistic Regression leads at 0.381 with calibration error 0.025, so it is the champion for deployment.

The RAG layer is specified as hybrid retrieval over official guidance, with citation checks and refusal rules. The alert service is specified as a period-aware rules engine with replaceable delivery adapters. Neither has measured field performance yet, because neither is deployed; the interfaces, rules and evaluation protocol are fixed so a later pilot can score them. The main limitation of the model is the label itself: a positive means documentation, not incidence, so reporting effort is the principal confounder and the main target for future work.

Keywords: Glanders, *Burkholderia mallei*, Risk Prediction, Gradient Boosting, Probability Calibration, SHAP, Retrieval-Augmented Generation, Citation Grounding, Alert Fatigue, Modular System Design, Animal Health Surveillance.

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Chapter 1

Introduction

1. Background

Glanders is a contagious and often fatal disease of horses, donkeys and mules caused by *Burkholderia mallei*. The organism has no environmental reservoir, and asymptomatic carriers remain infectious for long periods. Human infection is rare but carries a high case fatality rate and follows occupational contact. The disease is notifiable to the World Organisation for Animal Health and is classified as a potential bioterrorism agent.

The economic weight sits in low and middle income countries, where equids remain essential to transport, construction and agriculture. An equid is a member of the family *Equidae*: horses, donkeys (recorded here as asses) and mules. India held 5.50 lakh of them in the 2019 Livestock Census, made up of 3.42 lakh horses, 1.24 lakh asses and 0.84 lakh mules. That figure is the size of the susceptible population, not a count of sick animals: it says how many animals are exposed, not how many the disease has damaged. Most are held by households that depend on them for daily income, so the death of one working animal is a direct loss of livelihood.

The damage that is documented is smaller and better measured. A national programme screening 158,798 sera from 391 districts between 2019 and 2023 found 728 seropositive animals in 138 districts, roughly sixty per cent of them in Uttar Pradesh; earlier surveillance covering 2006 to 2018 reported 1,145 positives among 128,404 sera. That is under one per cent of the equid population, concentrated in a few states, and because glanders is notifiable and frequently fatal once clinical, each positive is acted on rather than monitored. Figure 1 separates those two quantities: the exposed equid population on one side, and documented occurrence events recovered from the literature on the other.

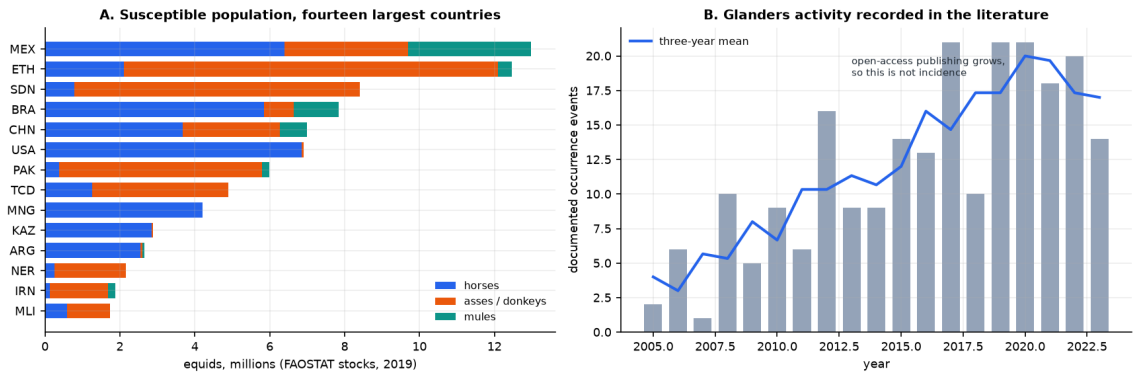


Figure 1: The two quantities behind the problem. Panel A gives the equid population of the fourteen largest countries, split by species, which sets the size of the exposed population. Panel B gives the occurrence events recovered from the literature each year, which sets the size of the label. The rise in panel B follows open-access publishing volume as much as disease, which is why the label is treated as documented activity throughout. Authors' plot from FAOSTAT livestock stocks [30] (panel A) and Europe PMC occurrence extraction [33] (panel B).

2. Problem Definition

A district veterinary officer faces a narrow question: is glanders likely to appear in this area in the coming year? The task is framed as supervised classification on a country-year panel: features observed in year t predict a binary label for year $t+1$. Answering it means combining the history of the disease nearby, the size of the susceptible population, and the conditions that permit transmission.

Surveillance reports what has already happened, and forecasting systems exist for several livestock diseases but not for glanders. A probability alone is still hard to use in the field: an owner needs to ask what to do next, and an officer needs answers that can be checked against official guidance. A warning that arrives for every area every month also teaches its recipients to ignore it.

The problem therefore has four parts, two concerned with producing the prediction and two with what happens after it.

1. **Data and label.** No public source records glanders occurrence completely, so occurrence must be reconstructed from the literature and joined to population and environmental covariates at a common resolution. The outcome is then rare, so accuracy is a poor summary statistic, and a negative observation needs an explicit meaning.
2. **Model selection under distribution shift.** A model fitted on some countries must be evaluated on countries it has never seen, because that is the deployment condition.
3. **Grounded question answering.** Officers and owners need to ask questions about control measures, reporting duties and next steps, and receive answers grounded in official guidance. Claims must cite retrieved passages; if the corpus does not support an answer, the system should refuse rather than invent one.
4. **Delivery without alert fatigue.** A probability is not an alert, so the operating point and the message rules must keep volume low enough that messages are read while the threshold stays defensible against the events caught.

3. Objectives

The project has three objectives.

1. **Prediction as a service.** Package a calibrated next-year risk model as a versioned artefact behind a REST interface, so any consumer can call it without knowing how it works.
2. **Grounded Q&A.** Build a retrieval-augmented layer that answers questions from official guidance, cites every claim, and refuses rather than guesses.
3. **Owner alerts.** Convert a calibrated probability into few enough messages that they are read, using threshold, suppression and deduplication rules.

All three share one requirement: each component must be independently testable and replaceable, and must not hardcode glanders into its core logic.

4. Scope

The study is scoped to country-year resolution. District-level modelling for India was envisaged but is not attempted, because the covariates could not be assembled from public sources within the project period. The RAG and alert layers do not depend on that resolution and can sit in front of a district-level model later without redesign. The model is a population-level tool: it does not diagnose an individual animal and does not replace laboratory confirmation. Limitations that shape the results are collected in Section 2..

Chapter 2

Literature Review and Gap Analysis

1. Epidemiology and Surveillance

Glanders is treated as a neglected zoonosis with high case fatality and a real occupational risk to handlers [1, 2]. A recent systematic review and meta-analysis consolidates seroprevalence estimates and identifies management practice and animal movement as the dominant drivers [3]; regional reviews document continued circulation in Brazil [4]. The disease persists where equid populations are large, animals move often and surveillance is weak. The first two are measurable from public data, while the third decides whether the disease is visible at all and is therefore a confounder rather than a covariate.

Indian surveillance is the strongest evidence base. Malik et al. described outbreaks reported between 2006 and 2011 [7], Singha et al. reported serological surveillance for 2015 to 2018 [6], and the national programme covering 2019 to 2023 produced a prevalence estimate with a risk analysis [5]. These establish where glanders occurs and which animal-level factors the risk analysis identified. None builds a predictive model, and none uses population or reanalysis covariates as predictors.

2. Existing Forecasting Systems in India

The closest operational system is NADRES v2, the National Animal Disease Referral Expert System run by ICAR-NIVEDI [34]. It combines outbreak records from AICRP centres with meteorological and remote-sensing covariates and issues district-level forewarnings about two months ahead, using a model family that overlaps with the one used here.

This project differs in four complementary ways. It is glanders-specific, where a national system must spread effort across many diseases. It uses only public data, so anyone can reproduce and audit it. It adds a question-answering layer over official guidance, which matters when the reader is an owner rather than a trained analyst. And it delivers to the owner directly, with rules that suppress routine noise. NADRES has institutional access this project lacks; this project spends that difference on local Q&A and owner delivery that a multi-disease platform does not attempt.

3. Glanders Risk Modelling and Related Machine Learning

Spatial work on glanders is descriptive: Fonseca-Rodríguez et al. applied spatiotemporal cluster detection to Brazilian data [8], which identifies where cases have already concentrated but does not predict which units will be positive in the next period.

Indian surveillance studies establish prevalence and animal-level factors but stop short of a forward model: Malik et al. on 2006–2011 outbreaks [7], Singha et al. on 2015–2018 serology [6], and the 2019–2023 national programme with risk analysis [5]. NADRES v2 issues multi-disease livestock forewarnings [34]; it is the closest operational comparator, but is not glanders-specific and does not deliver cited owner alerts from public data alone.

The only machine learning work specific to *B. mallei* is spectroscopic identification. Moawad et al. built

a Raman assay with learned classifiers [9], and Stöckel et al. reported spectroscopic detection of *B. mallei* and *B. pseudomallei* [10]. Both are laboratory diagnostic problems; neither produces a population risk estimate. Using published literature as structured occurrence data has precedent in epidemic intelligence [11, 12]; here extraction is the primary label source because official outbreak records are incomplete for many countries, so reporting bias is analysed rather than assumed away.

Because no glanders study compares model families on a shared panel, the evaluation design draws on method papers that are not disease-specific: how to compare classifiers under imbalance [17, 18], how to split data with spatial structure so country identity cannot leak [19], how to attribute predictions across features [16], and how to judge whether predicted probabilities are usable as probabilities [20, 21]. Those references justify the metrics and splits used in Chapter 4; they are not prior glanders risk models. Table 1 summarises both strands side by side.

Table 1: Comparative summary for the risk engine. Upper rows are glanders (or glanders-organism) studies; lower rows are method papers that guide multi-model evaluation.

Study	Approach	Problem addressed	Limitation
Fonseca-Rodríguez et al. (2019)	Spatiotemporal cluster detection on Brazilian glanders records	Where cases have already concentrated	Describes past clusters; no next-period prediction
Singha et al. (2020, 2026); Malik et al. (2012)	Serological surveillance and outbreak description in India	Prevalence and risk factors in equids	No forward predictive model on a covariate panel
NADRES v2 [34]	Multi-disease livestock forewarning	District-level early warning across diseases	Not glanders-specific; owner-facing cited delivery out of scope
Moawad et al. (2019); Stöckel et al. (2015)	Raman / spectroscopic classifiers for *B. mallei*	Laboratory identification of the organism	Diagnostic, not geographic risk
Saito and Rehmsmeier (2015); Davis and Goadrich (2006)	Precision–recall versus ROC analysis	How to score rare-event classifiers	Method only; no glanders application
Roberts et al. (2017)	Cross-validation for structured spatial / temporal data	Honest holdouts when units are correlated	Method only; guides leave-country-out here
Lundberg and Lee (2017)	SHAP feature attribution	Comparing what models use, across families	Attribution tool, not a disease model
Niculescu-Mizil and Caruana (2005); Huang et al. (2020)	Probability calibration measures	When a score can be read as a probability	Needed for alert thresholds; not glanders-specific

4. Question Answering and Retrieval-Augmented Generation

No glanders-specific question-answering system was found. The layer therefore follows general retrieval-augmented generation practice: retrieve passages, then answer only from those passages [22]. Dense embeddings [23], lexical retrieval [25] and reciprocal rank fusion [26] are the usual building blocks; none of them is a glanders product. What is missing is the application: a service that answers veterinary and owner questions from official control guidance, cites its sources, and refuses when the corpus does not support a claim. The risk score may be passed in as context for a question; the layer's job is still to answer the question, not to narrate the score. Table 2 lists the methods adapted for that layer.

Table 2: Related work for the RAG question-answering layer. There is no glanders-specific prior; rows are the methods adapted here.

Work	Approach	Problem addressed	Gap for this project
Lewis et al. (2020)	Retrieval-augmented generation	Knowledge-intensive NLP with evidence	Not wired to a veterinary Q&A workflow over statutory guidance
Reimers and Gurevych (2019)	Sentence embeddings for semantic search	Dense passage retrieval	Retrieval only; no citation or refusal protocol
Robertson and Zaragoza (2009)	BM25 lexical ranking	Keyword retrieval	Misses paraphrases; used here only as one hybrid channel
Cormack et al. (2009)	Reciprocal rank fusion	Combining ranked lists	Fusion method; not an end-to-end veterinary Q&A service

5. Alerting and Warning Systems

Related warning systems address pieces of delivery rather than a glanders risk-to-owner path: NADRES emphasises structured disease reporting, public flood warnings show how to name an area, a level and an action, early-warning evaluation work stresses receipt and usefulness rather than message send alone, software alert managers contribute grouping, silencing and routing, and standard public-warning formats support interchange. None of these is a complete substitute for the service designed here; Tables 3 and 4 summarise the lessons carried forward, with implementation detail later in Chapter 4 (Alerting System).

Table 3: Previous systems and the lessons applied in GlandersWatch.

Previous work	Main focus	Lesson for GlandersWatch
NADRES v2	Animal-disease surveillance and reporting	Use structured disease information and preserve traceability
Google / CWC flood warnings	Hazard communication to affected people	Name area, level and practical action, not only a probability
J-PAL Bihar early-warning study	Reach and usefulness of warnings	Evaluate receipt, understanding and response, not only sending
Prometheus Alertmanager	Grouping, silencing, inhibition and routing	Control repeats; separate decision from delivery
OASIS CAP	Standardised public-warning exchange	Keep alert records structured for possible external integration

6. Gaps and Contribution

Glanders epidemiology and laboratory identification are documented, yet a calibrated next-year population risk model on a public country-year panel is not. Retrieval-augmented generation is a mature method, but there is no glanders Q&A service that answers from official guidance, cites its sources and refuses when unsupported. Warning and alert-management systems exist in other domains, but none of them takes a calibrated glanders probability through hysteresis, suppression and deduplication to consented owner messages with an audit trail.

This project specifies a three-part platform behind fixed contracts: a risk engine that compares model families under temporal and leave-country-out validation, a RAG layer for grounded question answering, and an alert service that sends few enough messages to remain readable. Chapter 4 develops each part, while Chapter 5 reports measured results only for the risk engine.

Table 4: Difference between reviewed warning work and the proposed alert design.

Aspect	**Reviewed work provides**	**GlandersWatch proposes**
Input	Surveillance reports, hazard feeds or monitoring alerts	Calibrated glanders risk with area, period, model version and freshness
Period	Source-specific frequency	One period-aware contract for monthly and yearly records
Risk state	Warning or alert-management primitives	LOW / MEDIUM / HIGH with hysteresis and previous-state comparison
Notification control	Routing, silencing or distribution examples	Unchanged-level, cooldown, escalation, stale-data, mute, quiet-hour and daily-cap rules
Duplicate control	Grouping or deduplication in some systems	Separate protection for alert, recipient job and send attempt
Recipients / delivery	Domain- and platform-specific	Area subscriptions, preferences, queue fan-out and replaceable channel adapters
Audit	Varies	Reason-coded sent / skipped decisions and per-recipient delivery history

Chapter 3

System Architecture

1. What the Platform Has to Do

The risk engine answers whether glanders is likely to be documented in an area next period, while the platform also answers questions through the RAG layer and, when policy allows, delivers a short message to an owner. Those three jobs have different readers and different failure modes, so putting them in one process would couple alert delivery to LLM latency and make the risk model hard to test on its own. Figure 2 shows the three components and the prediction record that links them.

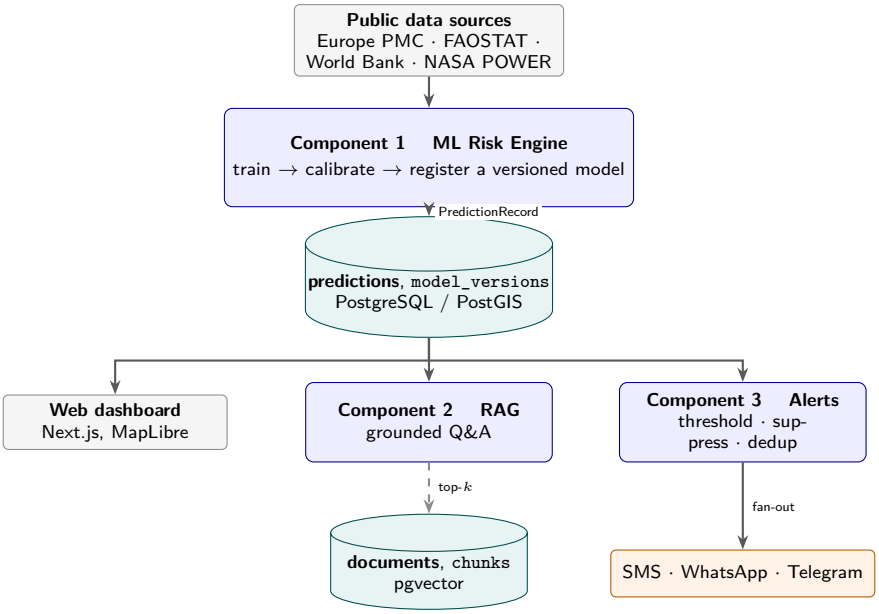


Figure 2: The platform decomposed into three components. The ML risk engine writes one `PredictionRecord` per area and period; the dashboard, the RAG layer and the alert service consume it independently. No component reads another component's tables, so any one of them can be replaced without touching the other two. Authors' diagram.

2. Design Principles

The split follows five practical rules.

Each component owns one job: the risk engine maps features to a calibrated probability, the RAG layer answers questions from the guidance corpus (optionally with the current prediction as context), and the alert service decides whether a prediction becomes a delivered message, with no component recalculating another's output.

Components exchange versioned schemas rather than shared tables or in-process calls, so each side is built and tested against a fixture and can be developed without waiting on the others.

The core stays disease-agnostic: a disease is a configuration record naming the label extractor, knowledge collection, templates and thresholds to use, so adding a second disease means new configuration and a retrain rather than a code fork.

Degradation is preferred over silence or invention: if the RAG layer is down, alerts still go out and chat returns a clear refusal, and if retrieval finds nothing above the relevance floor, the layer refuses and can attach the model's SHAP factors for the current prediction rather than invent guidance, so callers always get an explicit outcome.

Outputs stay auditable: a prediction records its model version, an answer records the documents it used, and an alert records the prediction and the rule outcomes behind the send-or-skip decision.

3. The Three Components

Table 5 states each component's job, interface and behaviour on failure.

Table 5: Component responsibilities, interfaces and failure behaviour.

Component	Responsibility	Interface	On failure
ML risk engine	Features to calibrated probability, with attribution	`POST /v1/predict`, `GET /v1/risk-map`	`503` with the champion id; the last good prediction is marked stale.
RAG layer	Answer questions from retrieved guidance (prediction optional as context)	`POST /v1/explain`, `POST /v1/chat`	Chat returns a refusal; alerts still send without a RAG answer.
Alert service	Prediction to a delivered, deduplicated message	`POST /v1/alerts/evaluate`	Alerts queue in Redis and drain on recovery; map is unaffected.

4. Component Contracts

A contract is a JSON schema plus its guarantees, and narrow contracts are what make a component replaceable. The risk engine emits a `PredictionRecord` with area, period, calibrated probability, risk level, model identifier, a freshness block and the top SHAP factors: the probability is calibrated on held-out data and can be read as a frequency, the risk level comes from the same probability through the threshold configuration so the two cannot contradict each other, and the SHAP factors come from the model itself so they remain available even if the RAG layer is down.

The RAG layer accepts an `ExplainRequest` (a natural-language question, with optional prediction context) and returns an `ExplainResponse` with an answer, resolved citations, a `grounded` flag and a refusal reason, with a refusal carrying no answer text. The alert service emits an `AlertEvent` with its identifier, area, period, both levels, source prediction, rule outcomes and recipient count. Components share a database but not tables, with spatial columns in PostGIS and embeddings in pgvector.

5. Modularity, Reuse and Status

The split is worth the overhead only if the parts are reusable: a second disease needs a new gazetteer, guidance collection and thresholds while the modelling code never names glanders and does not change, a second LLM provider is a configuration value because the RAG layer calls a single-method interface, and a second notification channel is an adapter plus a template. The pipeline is a scikit-learn compatible object [27] exported with a manifest, so any Python process can load it without going through the API.

This report separates what has been built from what has been specified: the risk engine is built and evaluated, every number in Chapter 5 is reproducible from the committed scripts, and serving, the RAG

layer and the alert service are specified but not deployed, so no measured performance is reported for them.

Chapter 4

Methodology

This chapter covers the three methods that make up the platform: the machine-learning risk model, the retrieval-augmented question-answering layer, and the alerting service.

1. Dataset Used

Four public data sources are used, and none had previously been applied to glanders risk modelling [33, 30, 32, 31]. Table 6 lists what each source contributes and at what size.

Table 6: The four data sources. Glanders events in Europe PMC were extracted by matching a country name and an event term within the same sentence, which yielded 263 events across 2000 to 2024.

Source	What it gives us	Size
Europe PMC (research papers)	Where and when glanders was reported	6,674 papers → 263 events
FAOSTAT	How many horses, donkeys, mules, cattle, goats and sheep each country has	68,449 records, 1961–2023
World Bank	Money, farming, population, internet access	10 indicators, 217 countries
NASA POWER	Temperature, rainfall, humidity, wind	6 measures, 211 locations

Why these four sources. They cover what decides whether a disease can appear at all: whether it has been there before, how many animals could catch it, and whether the climate suits it. World Bank data adds a rough measure of how well a country detects and reports disease. Pulling several kinds of data together is what makes this stronger than an analysis of one source. Extracted events rise after 2015, which reflects open-access publishing growth as much as incidence, so the label is treated as documented activity throughout.

The assembled panel. All four sources are keyed on ISO-3 country code and calendar year, so they join once names are reconciled. The joined panel starts at about **6,700 country-year rows and 60 columns** (identifiers, engineered candidate features, label fields and intermediate quantities) across 268 countries from 2000 to 2024. Preprocessing then reduces the row count: only equine-holding countries (maximum equid stock of at least 1,000), the years 2005 to 2023, and rows where next year's answer is observable are kept, which leaves 2,880 rows. Feature selection then reduces the column count further, from the engineered candidates down to the **8 input features** used for modelling, with 169 positive labels. That matrix is written as `model_matrix.csv` under `data/processed/` by the same loader the training script uses.

Training and testing split. The primary split is chronological, not a random 80/20 draw: models train on country-years from 2005 to 2017 (about 68% of the modelling window) and are tested on 2018 to 2023 (about 32%), so later years never enter fitting. Model selection also uses leave-country-out folds so every score is for an unseen country; Section 5. and Figures 3 and 4 show the full training and evaluation pipelines.

Preprocessing. Five steps, in this order.

1. **Name reconciliation.** Country names are matched across sources through `pycountry` with overrides for forms such as "Türkiye", so joins do not silently drop rows.
2. **Aggregate removal.** World Bank regional and global totals are dropped by requiring a three-character ISO code, so rows such as "World" never enter as countries.
3. **De-duplication.** Literature events are de-duplicated on `(iso3, year, record_id)`, so one paper cannot inflate a cell.
4. **Missing values.** Filled with the column median inside the training pipeline, with a missing-indicator flag per affected column. Missingness is low: 0.25% overall and at most 3.8% in any column.
5. **Absence handling.** A country-year is coded 0 only after the equine-stock filter, so a blank is never read as proof that disease was absent.

Feature engineering and selection. Candidate columns are built from information available in year t or earlier (disease history, livestock stocks, climate levels and anomalies, and country context). That widens the panel toward the sixty-column assembled table above. Selection then keeps eight numeric inputs by mean absolute SHAP on the training years alone (Section 2.), because marginal association and model attribution disagree about what matters. Section 3. describes only those eight retained features, the ones the models actually receive.

2. Dataset Analysis

We start with the label, then ask how each input relates to it and how much the inputs duplicate each other.

The answer key. For every country and year the answer is yes or no: was glanders documented the following year? Occurrence is extracted from Europe PMC by matching, within one sentence, a country name and an occurrence term (*outbreak*, *case*, *detected*, *seropositive*, *surveillance*). Because the label depends on publication it is termed *documented activity*: a negative may mean no disease or no report. Table 7 records the counts that follow from that definition.

Table 7: Numbers describing the answer key on the modelling matrix.

Measure	Value
Total rows	2,880
"Yes" rows (event next year)	169
"No" rows	2,711
Percentage "yes"	5.87%
Ratio of no to yes	about 16 to 1
Countries with at least one "yes"	55 of 160
Countries with "yes" in two or more years	30
Share of all "yes" in the top 5 countries	35.5%

What the distribution tells us. Three things follow from Table 7. First, yes is rare enough that always answering no would be right about 94% of the time, so accuracy is useless and we need scores built for rare events. Second, the disease comes back: 30 of the 55 positive countries were positive in two

or more years, so past activity is a fair input. Third, the yes answers cluster, with five countries holding 35.5% of them, and those five largely run publishing laboratories, which is why the target is documented activity rather than biological presence.

Statistical association. Before training, each numeric column is screened with Spearman's ρ (monotone association), mutual information (non-monotone dependence) and, for categories, Cramér's V. Variance inflation flags near-duplicates. The strongest epidemiological monotone signal is `years_since_last`: more years since the last report means lower next-year risk. Columns such as longitude can rank high on marginal association while ranking low under model attribution, because they stand in for which region publishes rather than for transmission itself.

Where the two importance measures disagree. Selection therefore uses SHAP attribution on the training years (2005–2017) rather than marginal correlation: mutual information and Spearman can favour reporting proxies, while mean absolute SHAP asks how much a feature adds once the others are already in the model [16]. The 42 numeric candidates are ranked on that training window and the first eight are kept; the exported matrix contains exactly those eight columns so the artefact on disk matches the model behind the results. The retained set carries 49% of total attribution across all candidates. Chapter 5 shows what that compression costs in transfer.

Table 8: The eight features the reported models use, in rank order by mean absolute SHAP on the training years.

Rank	Feature	Unit as measured	Mean $\lvert\phi\rvert$
1	`pop_total`	persons	1.035
2	`years_since_last`	years (99 = none)	0.648
3	`temp_mean_anom`	°C from baseline	0.505
4	`internet_pct`	% of population	0.449
5	`humidity_anom`	% RH from baseline	0.382
6	`precip_cv`	ratio (sd/mean)	0.371
7	`log_livestock`	ln(1 + LSU)	0.364
8	`goats`	head	0.362

What we removed, and why. `equids` is dropped because it is exactly the sum of horses, asses and mules. Near-duplicate pairs and columns with no measurable label relationship are discarded. No principal component analysis is used: with eight retained features and 2,880 rows the binding constraint is signal rather than dimensionality.

3. Features

The models are given exactly the eight features in Table 8, all computed from year t or earlier so they can predict the year-t+1 label. In rank order they are total population, years since the last documented event (99 if none in the window), mean-temperature anomaly relative to that country's own expanding baseline, internet penetration, humidity anomaly, precipitation coefficient of variation, log total livestock (LSU), and goat headcount. Earlier engineering built a wider candidate pool from history, stocks, climate and country context; those candidates are not passed to the fitted models once SHAP selection has chosen this set of eight.

4. Models

Six families are compared, including a deliberately simple baseline. Table 9 lists each family and what it contributes.

Forest models train trees in parallel on random subsets and average them, which reduces variance. Boosted models train trees in sequence, each fitting the error left by the last, which reduces bias. Neural networks are left out because with 169 positives and eight features a network has more parameters than

Table 9: The six models and what each is for.

Model	Family	What it does
Logistic regression	linear	Adds the inputs with fixed weights and maps the total to a 0–1 score; the honest baseline, with balanced class weights.
Decision tree	tree	A shallow flowchart of yes/no splits that a non-specialist can read; weak on its own, which is why it is included.
Random forest	bagged trees	Many trees trained on random subsets, then averaged [15].
Extra trees	bagged trees	Like random forest, with extra split randomisation [15].
XGBoost	boosted trees	Trees trained in sequence, each fitting residual error [13].
LightGBM	boosted trees	A faster boosted variant with leaf-wise growth [14].

examples and would memorise the training period. Tuning uses grouped cross-validation on country. We do not stack the families into a meta-learner: the gap between scores is smaller than the uncertainty on any single score, so that effort goes into calibration instead.

How the final model is chosen. On leave-country-out precision-recall AUC, because deployment means an unseen country rather than an unseen year. Chapter 5 reports that ranking: logistic regression leads under that criterion even when a tree ensemble leads on the easier temporal holdout.

5. Training and Evaluation

Several measures keep anything from the test period out of training. Figure 3 shows the steps from the four sources through the chronological split; Figure 4 shows how one trained model is scored.

Three schemes answer different questions. A temporal holdout trains on 2005 to 2017 (about 68% of rows) and tests on 2018 to 2023 (about 32%), asking whether the model can be used in following years. Leave-country-out uses five folds grouped by country, so every prediction is for an unseen country; that is the harsher test and the one that governs selection. Repeated stratified cross-validation (five folds, three repeats) is a stability check. A random row split would let the model memorise country identity, which is why splits are grouped [19].

Precision-recall AUC is the primary metric: at 5.87% prevalence, ROC-AUC is dominated by the true-negative mass and overstates performance [17, 18], so ROC-AUC is reported only for comparability. Brier score and expected calibration error check whether predicted probabilities can support a threshold-based alert [20, 21]. Precision, recall, specificity, F1 and Matthews correlation use a threshold tuned on training data only, with bootstrap percentile intervals over one thousand resamples.

Leakage controls. Features use year t or earlier and only the target uses $t+1$; the split is chronological; hyperparameters are tuned inside training folds; calibration is fitted on training folds only; the alert threshold is fixed on training; and imputation values and missing indicators are generated inside the pipeline.

Why PR-AUC, calibration and SHAP. PR-AUC asks how quickly ranking the riskiest rows surfaces real events; a method with no skill scores the prevalence (0.0587 here). Calibration groups predictions into buckets and compares each bucket's average prediction with how often the event happened; the averaged gap is expected calibration error. SHAP assigns each input a fair share of one prediction by averaging marginal contributions over feature orderings [16]: the values sum to the gap between the base rate and that prediction, which is why the same numbers support both global feature ranking and the per-prediction factor list the risk engine returns.

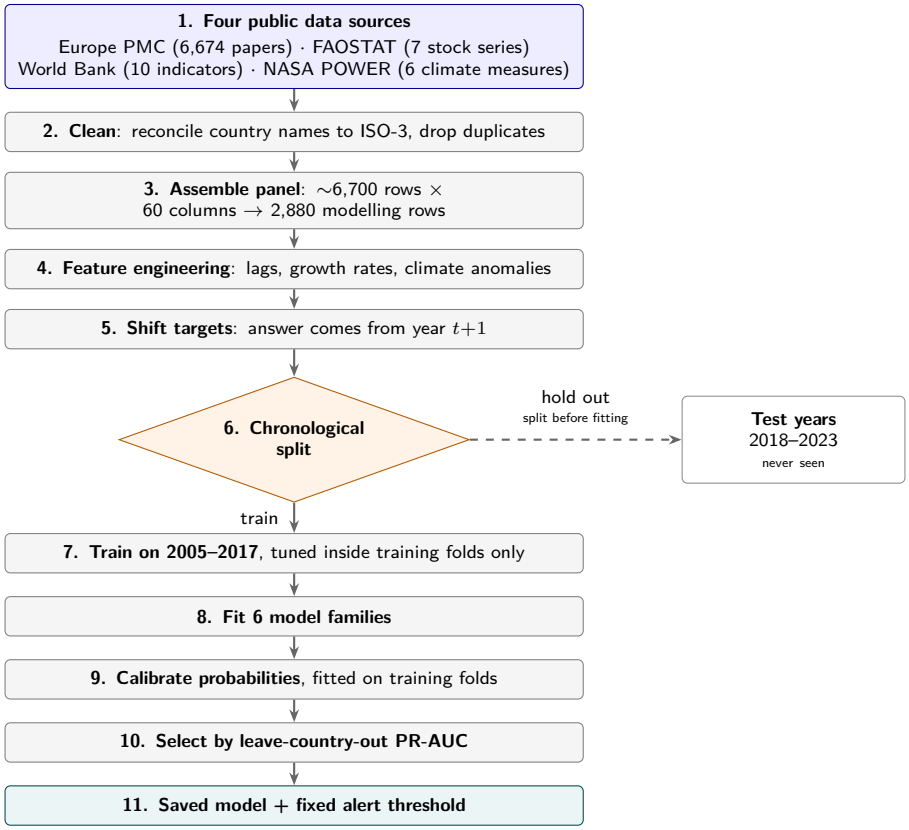


Figure 3: Training pipeline. The chronological split sits before any model sees the data, so the 2018–2023 years never influence training. Tuning and calibration both happen strictly inside the training years. Authors' diagram.

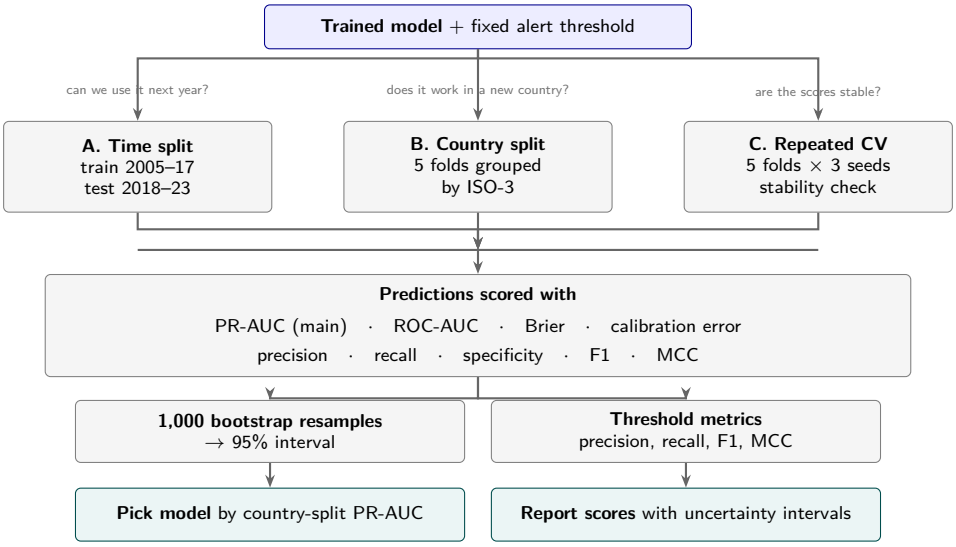


Figure 4: Evaluation pipeline. One trained model is scored three different ways, each answering a different question, then reported with uncertainty intervals. Authors' diagram.

6. Retrieval-Augmented Question Answering

6.1 The Problem This Layer Solves

A calibrated probability of 0.24 is useful to a veterinary officer and opaque to most equine owners, yet both still have questions that the risk score does not answer: what the National Action Plan requires, how movement control works, what to do if an animal looks sick, and whether a claim is still current guidance. Asking a large language model directly is the obvious shortcut and a real hazard, since a model asked about glanders may answer from outdated training data, conflate glanders with melioidosis, or invent a treatment protocol, and for a zoonosis under statutory control an invented instruction is worse than no answer. Retrieval-augmented generation [22] restricts what the model may use: it receives a small set of passages from official guidance and, when relevant, the current prediction's numbers, and must answer only from those passages, so each claim traces to text the reader can check.

The goal is narrow and testable: answers a veterinary officer would accept as consistent with the National Action Plan, and a refusal rather than a guess when the corpus does not support one. Fine-tuning would give fluent text without citations, no clean line between guidance and the base model's priors, and knowledge frozen at training time, whereas retrieval keeps knowledge in a corpus that a re-ingestion can update, makes every answer auditable, and allows a refusal path, at the cost of latency and possible retrieval failure.

6.2 Knowledge Base

The corpus is restricted to authoritative sources so that answers can be checked against them, and each document carries an authority tier used as a tie-breaker during retrieval. Table 10 shows the tiers.

Table 10: Knowledge base, by authority tier.

Source	Tier	Use
DAHD National Action Plan for Glanders, 2025 [29]	1	Case definitions, surveillance obligations, movement control, quarantine, notification duties.
WOAH and WHO disease information sheets	1	Disease biology, transmission, zoonotic risk, official status.
ICAR and national laboratory guidance	2	Diagnostic sampling, laboratory referral.
Open-access publications	3	Epidemiology, risk factors, regional situation.

Tier 3 may inform an answer but never overrides tier 1 on a control instruction. Conflicts resolve to the higher tier and are logged, because a silent contradiction between sources is the failure this layer exists to prevent.

6.3 Ingestion, Embedding and Retrieval

Ingestion runs offline as a job and stores each chunk with enough metadata to be cited and filtered. Chunking is section-aware rather than fixed-width, because a fixed window cuts control instructions in half and lets the model complete a truncated sentence. The splitter respects headings first, then paragraphs, falling back to a sliding window only past the ceiling. Chunks target 400 to 700 tokens with 15 per cent overlap inside a section, and sections under 80 tokens merge into the parent. The full section path is retained, for example `NAP-2025 > 5.2 Movement control`, so a citation points to a subsection rather than a 90-page document.

Chunks are embedded with a compact sentence-transformer [23] producing 384-dimensional vectors, with a multilingual variant for Hindi because an English-only embedding degrades on Devanagari. Vectors live in pgvector beside the relational tables, so re-ingestion and prediction share one transaction boundary. The index is HNSW with cosine distance [24]; the reranker corrects recall at the head of the list, and the index keeps query cost flat as the corpus grows. Figure 5 shows the offline ingestion path above the per-request retrieval and generation path.

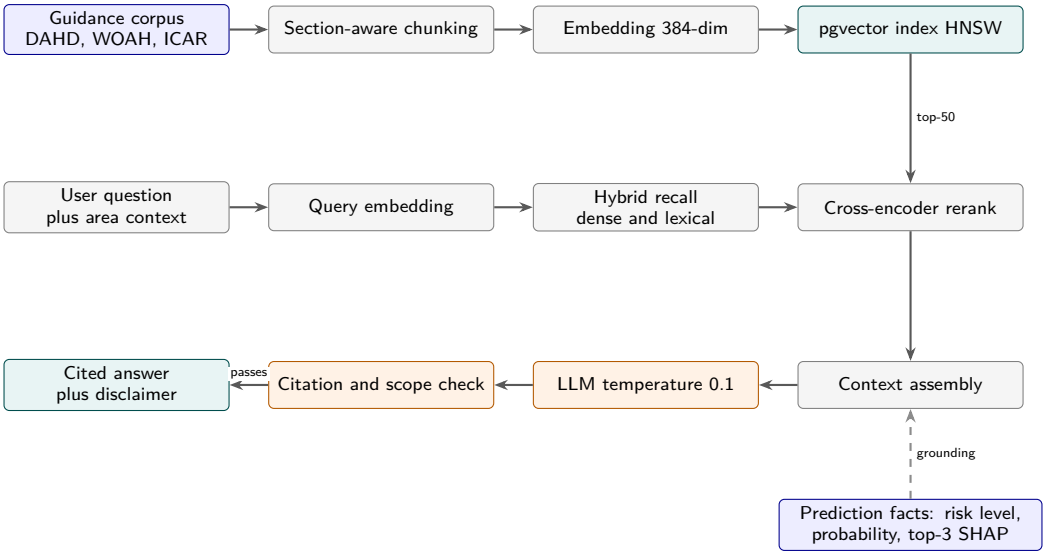


Figure 5: The RAG layer. Ingestion (top row) is an offline job; retrieval, assembly and generation (below) run per request. Predicted risk and SHAP factors are injected as structured facts, so the answer explains this forecast rather than glanders in general. The citation check sits after generation and can return a refusal instead of an unsupported answer. Authors' diagram.

Retrieval is hybrid, because dense and lexical search [25] fail differently here. A dense query for "movement control" may miss a passage using "restriction of equid movement", where a lexical match succeeds; conversely a lexical query for a rare acronym retrieves noise a dense query ignores. Both lists are fused by reciprocal rank fusion [26], which combines rankings without requiring comparable scores:

$$\mathrm{RRF}(d) = \sum_{i \in \{\mathrm{dense},\mathrm{lexical}\}} \frac{1}{k + \mathrm{rank}_i(d)}, \qquad k = 60$$

The fused top fifty are reranked by a cross-encoder scoring each passage against the question jointly. It is far slower per pair than an embedding dot product, which is why it runs on fifty candidates rather than the corpus, and the top five become the context. Table 11 records the stage-wise settings.

Table 11: Retrieval parameters.

Stage	Setting	Reason
Dense recall	top 30, cosine	Semantic paraphrase coverage
Lexical recall	top 30, BM25	Exact terms, acronyms, section titles
Fusion	reciprocal rank, $k = 60$	Scale-free combination of two rankings
Rerank	cross-encoder, top 50 to 5	Precision at the head of the list
Relevance floor	rerank score below 0.30 discarded	Prevents padding the context with weak passages

The relevance floor drives the refusal path: if fewer than two passages clear it, the layer returns a refusal with the SHAP factors, because marginal passages produce a confident answer built on nothing.

6.4 Context, Generation and Guardrails

The prompt has three parts in fixed order: a system instruction, the retrieved passages with identifiers, and (when supplied) structured prediction facts for the area under discussion. The prediction block is context for questions about a specific forecast; it is not required for general guidance questions. Facts are injected as structured data so the model cannot rewrite an association as a cause. The instruction states four constraints: answer only from the passages, cite every claim, never diagnose an individual animal, and make clear when a population-level forecast is being discussed.

Generation uses temperature 0.1 and a bounded output length so that two officers asking the same question about the same area tend to get the same instruction. The provider sits behind a single-method

interface, so one retrieval, grounding and citation path can serve a local Ollama model or a hosted API; switching model cannot silently disable the guardrails.

The layer is designed to refuse, and four checks can terminate a response. Table 12 states each check and what happens when it fails.

Table 12: Guardrails and their failure behaviour.

Check	What it verifies	On failure
Retrieval floor	At least two passages above the relevance threshold	Refusal with the top SHAP factors, which need no retrieval
Citation resolution	Every cited identifier exists in the retrieved set	Answer rejected and regenerated once, then refusal
Diagnosis scope	No individual-animal diagnosis or treatment prescription	Response rewritten without the clinical claim, and the attempt logged
Out-of-scope	Question is about glanders control, not unrelated topics	Refusal stating what the layer covers

The citation check is mechanical: the answer is parsed for identifiers and each is looked up in the retrieved set, and one that does not resolve means the model cited a document it was not given, which is a hallucination whether or not the sentence is true. Asking a model to be accurate is not a control; checking its output is.

6.5 Evaluation

Five quantities are defined below, following the project proposal. None is reported as a measurement, because the layer has not been run against the corpus; measuring them is future work (Chapter 6).

Retrieval quality. Recall@k, the fraction of hand-built questions whose known relevant passage appears in the top k results, at $k = 5$, with mean reciprocal rank, which rewards returning it near the top rather than merely inside the cut-off. This isolates retrieval from generation, since no generation quality helps if the right passage is never retrieved.

Faithfulness. The fraction of claims the retrieved context supports, scored by a judge model and reviewed by hand on a sample; the RAGAS-style metric that catches a model citing a passage while saying something the passage does not say.

Answer relevance. Whether the answer addresses the question asked rather than reciting correct but tangential guidance, scored on the same sample, since an answer can be grounded and still unhelpful.

Citation precision. The fraction of citations resolving to a passage that supports the sentence, distinct from resolution, which only checks existence.

Refusal behaviour. The refusal rate on out-of-scope questions and the false refusal rate in scope: a layer that never refuses is unsafe and one that always refuses is useless.

Deployment targets, so the claim is falsifiable: recall@5 above 0.85, faithfulness and answer relevance above 0.90 and 0.85, citation precision above 0.95, a false refusal rate below 10 per cent, and p95 latency under two seconds.

7. Alerting System

7.1 Purpose and Scope

The GlandersWatch alerting system converts a machine-learning risk prediction into a controlled notification for registered recipients. The risk engine estimates the probability of documented Glanders occurrence for an area and period; the alert service decides whether to notify, who should receive the message and how delivery is managed. A prediction is not itself an alert: every new prediction is evaluated, but a notification is generated only when the configured policy requires it.

The design is period-aware and uses one workflow for monthly and yearly predictions. Monthly prediction is future work for the ML component; the currently evaluated model is country-year based. The alert service is designed to accept monthly records once a monthly model, target and data pipeline have been

developed and validated. Its output is a population-level warning, not an individual-animal diagnosis or a replacement for veterinary examination and laboratory testing.

Related warning systems, design lessons and the comparison with this service are reviewed in Chapter 2 (Tables 3 and 4). This section specifies the architecture and rules that implement that design.

7.2 Proposed End-to-End Architecture

Figure 6 shows the supplied end-to-end architecture. The flow begins with user registration and prediction generation, then proceeds through evaluation, rules, alert creation, fan-out and notification. The data layer stores predictions, subscriptions, period-specific alert state, alerts and delivery logs. The approval gate is optional and may hold selected HIGH alerts for veterinary review; its use requires agreement with project and veterinary stakeholders.

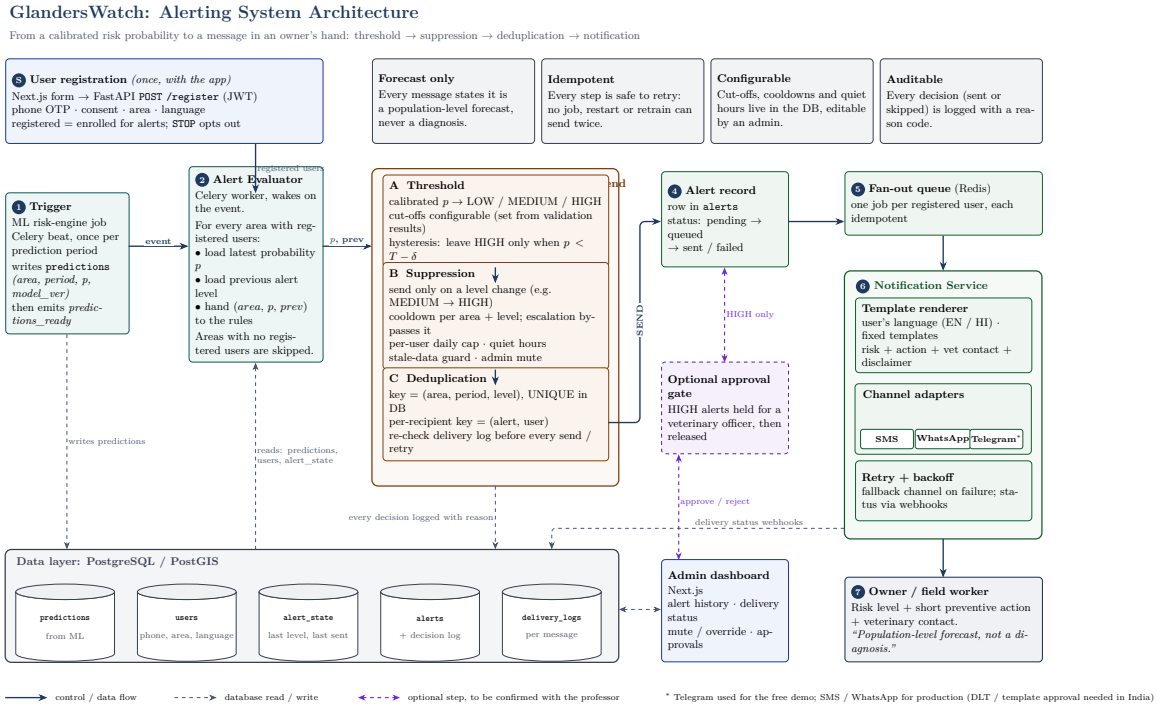


Figure 6: End-to-end GlandersWatch alert pipeline. The period field supports monthly or yearly predictions. Authors' diagram.

Prediction Trigger and Alert Evaluator

After the risk engine stores a prediction, it publishes `predictions_ready`, which tells the evaluator that a prediction batch is available without implying that a message must be sent. The evaluator loads the new prediction, previous state for the same area and frequency, active subscriptions and applicable configuration, then passes these inputs to the rules core; it does not train the model or calculate the probability.

Rules, Alert Record and Fan-Out

The rules engine classifies the probability, checks suppression conditions and protects alert creation against duplicates. If the decision is SEND, one logical alert record is created with its area, period, level, source prediction, reason and status. This record is separate from delivery jobs: one alert for Area A may generate 41 jobs for 40 owners and one veterinary officer.

Fan-out obtains eligible subscriptions and creates one job per subscription. Redis with Celery is a proposed production option: Redis brokers background work and Celery workers execute it. The queue is not the permanent alert-history store; PostgreSQL is proposed for durable predictions, states, alerts and delivery logs.

Notification Service, Data and Administration

For each job, the notification service checks recipient preferences and delivery restrictions, renders an approved template in the recipient's language and submits it through a channel adapter. The design allows SMS, WhatsApp and Telegram, with email or webhook adapters as extensions. Messages should state the area, period and risk level, provide approved action guidance and veterinary contact information where available, and clarify that the result is a population-level forecast, not a diagnosis.

Temporary provider failures may be retried with backoff, while permanent failures such as invalid destinations are recorded without endless retries. PostgreSQL stores predictions, users/subscriptions, alert state, alert records and delivery logs, with PostGIS available if geographic queries are needed, and the admin dashboard exposes alert history, delivery status, mute controls and any approval workflow under permission-controlled, logged administrative actions.

Notification Content and Recipient Action

A useful warning should be short enough to read on a phone but contain enough context for the recipient to understand why it was sent. Each message should identify the affected area, prediction period, risk category, issue date and recommended next step. It should avoid technical model terminology and should not present the probability as the chance that a particular animal is infected. The message should also identify the responsible veterinary contact or official information source, where one is available. The language and channel must follow the recipient's saved preferences and consent.

An illustrative message for a HIGH-risk transition is:

> **GlandersWatch risk alert – Area A**
> **Period:** March 2027
> **Risk level:** HIGH
> The model indicates an elevated area-level risk of documented Glanders occurrence for this period. This is an early-warning signal, not a diagnosis of any animal. Avoid moving or mixing animals suspected of illness, observe for unusual signs and contact your local veterinary officer for guidance. Do not handle suspected cases without appropriate advice.
> *For assessment and reporting, contact: [local veterinary contact].*

The exact action wording should be approved by veterinary authorities before deployment and adapted to local language and operating procedures. A message should not disclose another user's contact details or unnecessary personal information. If no verified local contact is configured, the system should direct recipients to an approved general veterinary service rather than inventing a phone number. Message templates should be versioned so the content used for each alert can be reconstructed during an audit.

7.3 Period-Aware Prediction Contract

The prediction must identify its frequency and exact period. A conceptual monthly record is shown below; field names are illustrative and can be aligned with the final API schema.

```
{
  "area_id": "AREA-A", "period_type": "month",
  "period_id": "2027-03",
  "period_start": "2027-03-01", "period_end": "2027-03-31",
  "probability": 0.34, "model_version": "gw-model-v1.2",
  "data_freshness": {"passes": true}
}
```

For yearly records, `period_type` is `year` and the period ID identifies the year. The same evaluator and rules core process either type. State is stored separately by area and frequency: a March 2027 monthly result must not overwrite Area A's 2027 yearly state.

A period-aware logical alert key is:

$$K_{alert} = (\text{area_id}, \text{period_type}, \text{period_id}, \text{risk_level}).$$

This prevents a monthly alert from colliding with a yearly alert for the same area. Cooldowns, freshness requirements and thresholds may also be configured by frequency; monthly settings should not simply inherit yearly values without validation.

Table 13: Monthly and yearly operation in the same alert service.

Aspect	Monthly	Yearly
Period identity	Area + month ID	Area + year ID
Trigger	Monthly prediction-ready event	Yearly prediction-ready event
Previous state	Monthly stream state	Yearly stream state
Deduplication	Area + month + level	Area + year + level
Policy	Monthly thresholds/cooldown as validated	Yearly thresholds/cooldown as validated

Table 13 summarises how the two frequencies stay distinct inside one service.

7.4 Rules Engine: Thresholding, Suppression and Deduplication

Thresholding and Hysteresis

The calibrated probability is mapped to LOW, MEDIUM or HIGH. Numerical boundaries must be selected from model validation and the trade-off between false alerts and missed events; 0.5 must not be assumed to be the correct HIGH threshold. Table 14 gives the intended meaning of each level.

Table 14: Meaning of the proposed alert levels.

Level	Interpretation	Example guidance
LOW	Below configured MEDIUM boundary.	Routine monitoring; normally no risk-change message.
MEDIUM	Between MEDIUM and HIGH boundaries.	Monitor animals, prepare and keep veterinary contacts available.
HIGH	At or above configured HIGH boundary.	Follow approved precautions, avoid unnecessary movement of suspected animals and contact a veterinarian/authority.

Hysteresis reduces frequent switching near a boundary. An area enters HIGH when $p \geq T_H$ and leaves HIGH only when $p < T_H - \delta$, where T_H is the HIGH entry threshold and δ is a configurable margin. For illustration, if $T_H = 0.30$ and $\delta = 0.05$, a probability of 0.32 enters HIGH; a later value of 0.29 remains HIGH; the area leaves only below 0.25. The margin must be validated for the relevant prediction frequency.

Suppression

Suppression decides whether a notification is necessary; it does not delete or ignore the prediction. The evaluator still processes each new prediction. The rules are:

1. **Unchanged level:** normally notify on a risk-level transition, not every period. MEDIUM in consecutive months does not automatically create repeated messages.
2. **Cooldown:** hold a repeat of the configured type/level within a time window; for example, a 48-hour cooldown can hold a repeat inside that period.
3. **Escalation bypass:** a permitted upward change, such as MEDIUM to HIGH, can bypass ordinary cooldown so a more serious warning is not delayed.
4. **Stale-data guard:** hold or skip an ordinary alert if freshness checks fail.
5. **Area mute:** an authorised administrator may temporarily silence an area; record the action and reason.

6. **Quiet hours:** defer delivery during the recipient's local quiet period, subject to the urgency policy.
7. **Daily recipient cap:** limit messages to one recipient in a day, including when they subscribe to several areas.

Quiet hours and daily caps are checked by the dispatcher at delivery time because they depend on local time and other messages sent to that recipient. Stale data, area mute and risk-state transition are checked earlier. An optional all-clear/de-escalation message may be sent when risk falls, but it must not imply that an individual animal is disease-free.

Deduplication

Deduplication protects against replayed events, worker restarts and concurrent processing. It is distinct from suppression: suppression decides whether a message is warranted; deduplication prevents the same logical operation from being created or processed twice.

1. **Alert level:** unique key `(area, period_type, period_id, level)` prevents duplicate logical alert records.
2. **Recipient level:** unique pair `(alert_id, subscription_id)` prevents duplicate jobs for one subscription.
3. **Send level:** check delivery logs before retrying and use provider idempotency keys where available, to manage uncertain timeouts.

The first two protections should use database unique constraints, not only application checks. A local database cannot guarantee exactly-once delivery to an external provider; callbacks, status queries and reconciliation are needed where supported.

7.5 Modular Architecture: Core, Ports and Adapters

Figure 7 shows the internal modular structure. The alert engine core contains domain rules and does not directly depend on a database, web framework or messaging SDK. Four ports define what the core needs; adapters implement them for a demo or production stack.

RecipientProvider. Finds active subscriptions for an area. A demo may read a JSON file; production may query user/subscription tables or a registration service. Registration should capture verified contact details, area, language, consent and opt-out status.

RuleConfig. Supplies thresholds, hysteresis, cooldowns, daily caps, quiet hours, mute and approval settings. It may use `config.json` in a demo and database/admin UI in production. Monthly and yearly policies can differ after validation.

StateStore. Reads previous state and saves alerts, decision reasons and delivery outcomes. In-memory storage or SQLite may support a demo; PostgreSQL is proposed for durable operation. Redis may broker jobs but should not be the sole alert-history store.

Notifier. Submits messages through a common interface while hiding provider-specific APIs and authentication. Demo options include a simulated channel or JSONL outbox; possible adapters include SMS, WhatsApp, Telegram, email, webhook and CAP export. Production SMS/WhatsApp require provider onboarding and applicable template approvals.

This Ports-and-Adapters (Hexagonal) structure separates business rules from technology choices. Replacing SQLite with PostgreSQL or Telegram with an approved SMS provider changes an adapter, not the threshold and suppression logic. The core can initially run as a Python library and later be hosted behind an API; this design does not require every component to be a separate microservice.

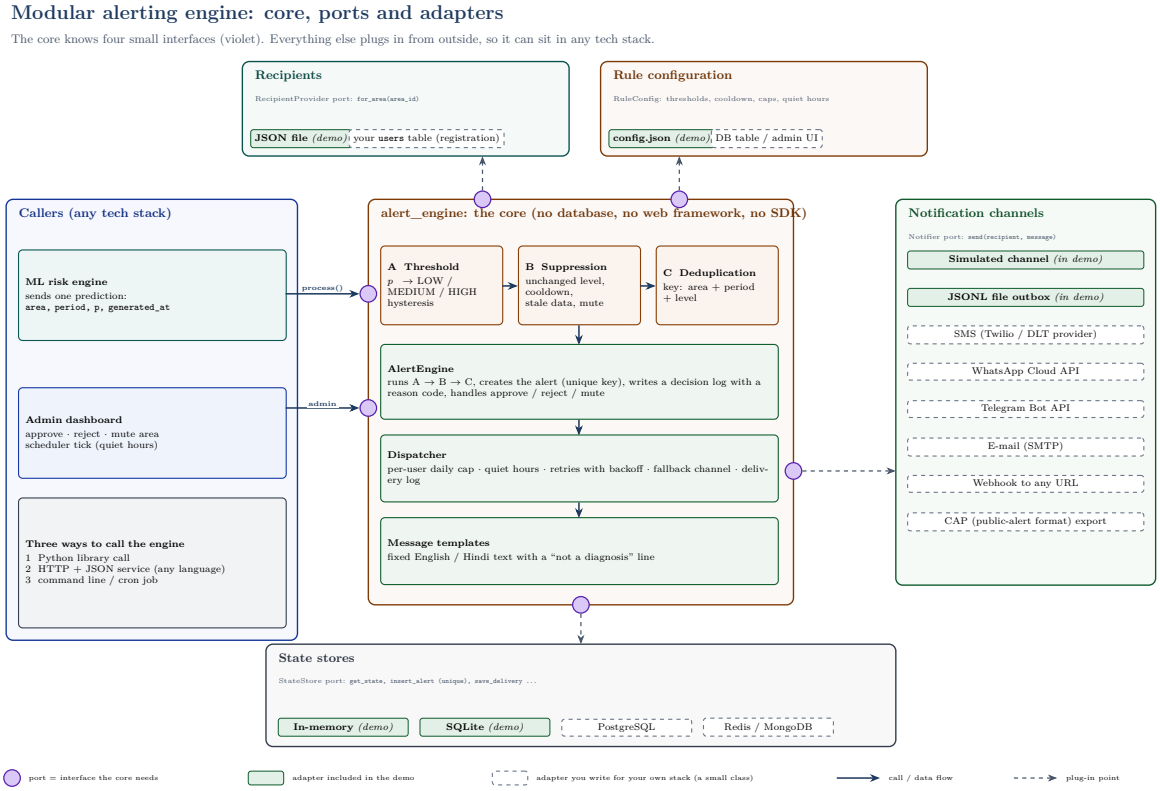


Figure 7: Modular alerting engine: rules core, four ports and replaceable adapters. Authors' diagram.

Table 15: Illustrative monthly state sequence for Area A.

Month	Probability	New level	Previous	Decision
Jan	0.22	MEDIUM	LOW	Send transition if checks pass.
Feb	0.24	MEDIUM	MEDIUM	Suppress repeated risk-change message.
Mar	0.34	HIGH	MEDIUM	Send escalation if checks pass.
Apr	0.29	HIGH	HIGH	Remain HIGH by hysteresis; no repeated transition.
May	0.23	HIGH	HIGH	Remain HIGH; value is not below 0.25.
Jun	0.24	HIGH	HIGH	Remain HIGH; no new transition.
Jul	0.20	MEDIUM	HIGH	Optional de-escalation message.

7.6 Example Flow: Monthly and Yearly Predictions

The following monthly example uses fictional values only. Assume MEDIUM begins at 0.15, HIGH entry is 0.30, hysteresis is 0.05, and Area A has 40 owner subscriptions plus one veterinary officer. Table 15 walks through seven months of that sequence.

For March, the evaluator consumes the event, loads the prediction, previous monthly state and subscriptions. The rules classify 0.34 as HIGH and detect MEDIUM-to-HIGH escalation. Freshness, mute and cooldown policies are checked; an allowed escalation bypass may override ordinary cooldown. The period-aware alert key is checked, and one alert record is created if it does not exist. Fan-out creates 41 recipient jobs, protected by the recipient unique key. The dispatcher applies quiet hours and daily caps, renders each recipient's language template, sends through the selected adapter and records each outcome.

In April, 0.29 does not cause the area to leave HIGH because the exit boundary is below 0.25. In July, 0.20 is below that boundary and the state can de-escalate to MEDIUM. The same rules apply to yearly predictions: if the 2027 yearly state is MEDIUM and the 2028 yearly state is HIGH, the engine evaluates the transition using the yearly state and a key such as `(Area A, year, 2028, HIGH)`. Monthly and yearly states remain separate.

7.7 Reliability, Audit and Evaluation

Every decision, including a skipped alert, should be logged with a reason code such as `UNCHANGED_LEVEL`, `STALE_DATA`, `AREA_MUTED`, `COOLDOWN`, `DUPLICATE_ALERT` or `QUIET_HOURS`. The alert record explains why a logical alert was created; delivery logs record the outcome for each recipient.

Transient failures, such as temporary timeouts, may be retried with exponential backoff and jitter. Permanent failures, such as invalid contact details, should not be retried indefinitely. If a provider may have accepted a message but the application timed out, check the delivery log and provider status before retrying. Provider acceptance and confirmed recipient delivery should be treated as different statuses when delivery receipts are available.

Evaluation should include: (i) alert precision/false-alert rate; (ii) event coverage/missed-event rate; (iii) notification noise per recipient per month and year; (iv) delivery success by channel; (v) time from `predictions_ready` to decision and provider acceptance; and (vi) idempotency tests using event replay, worker restart, concurrent evaluation and provider-timeout simulation. These are proposed measures, not measured production results.

7.8 Limitations and Summary

Alert-specific limits (no production delivery yet, and a send that is not proof of reading) are stated with the platform-wide list in Section 2.. In short, the service is a period-aware rules core that turns a calibrated prediction into a controlled notification: thresholding and hysteresis set the level, suppression and deduplication keep the volume readable, and ports keep storage and channels replaceable.

Chapter 5

Results and Discussion

1. The Risk Model

1.1 Dataset Characteristics

The modelling matrix holds 2,880 country-year observations, of which 169 are positive: a prevalence of 5.87% and an imbalance of 16.0 to one. 55 of 160 countries record at least one positive, 30 of them in two or more years, and the top five hold 35.5% of all positives.

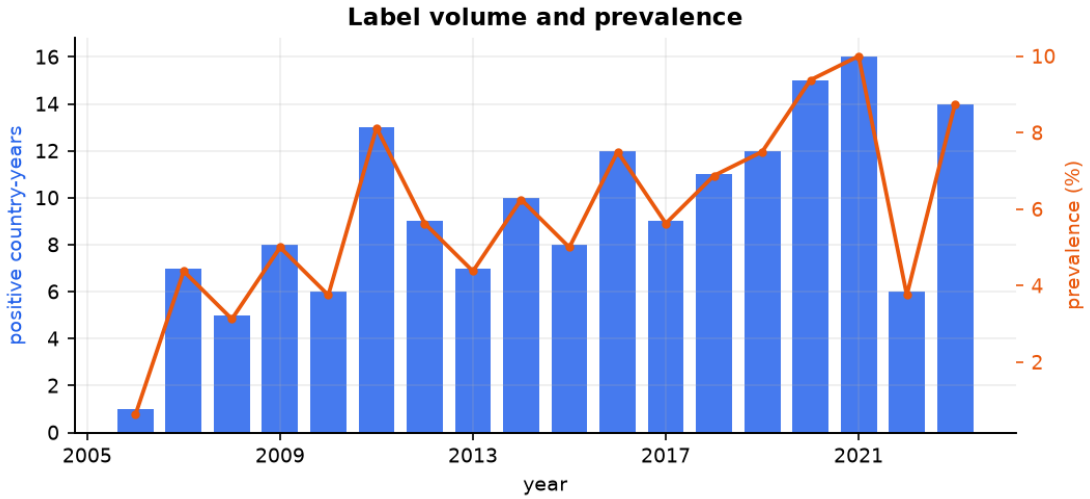


Figure 8: Annual positive country-year counts (bars) against prevalence on the secondary axis (line). Authors' plot from the assembled modelling matrix (Europe PMC labels [33]).

Three properties shape the design. The imbalance, visible as the gap between the count and prevalence axes in Figure 8, rules out accuracy as a metric. The persistence in Figure 9, with 30 of 55 positive countries recurring, justifies historical features. The regional concentration in Figure 10 is the first indication that the label tracks reporting capacity, since those countries operate reference laboratories.

1.2 Model Comparison

Table 16 reports discrimination under all three schemes.

Two results need separating. On the forward temporal holdout the best discrimination comes from Random forest at 0.495 against a base rate of 5.87%, roughly a 8-fold lift. Under leave-country-out validation the ranking inverts: Random forest falls to 0.322, while logistic regression holds 0.381 and becomes the strongest model on the criterion that governs selection. Its temporal score is 0.461, about a 8-fold lift.

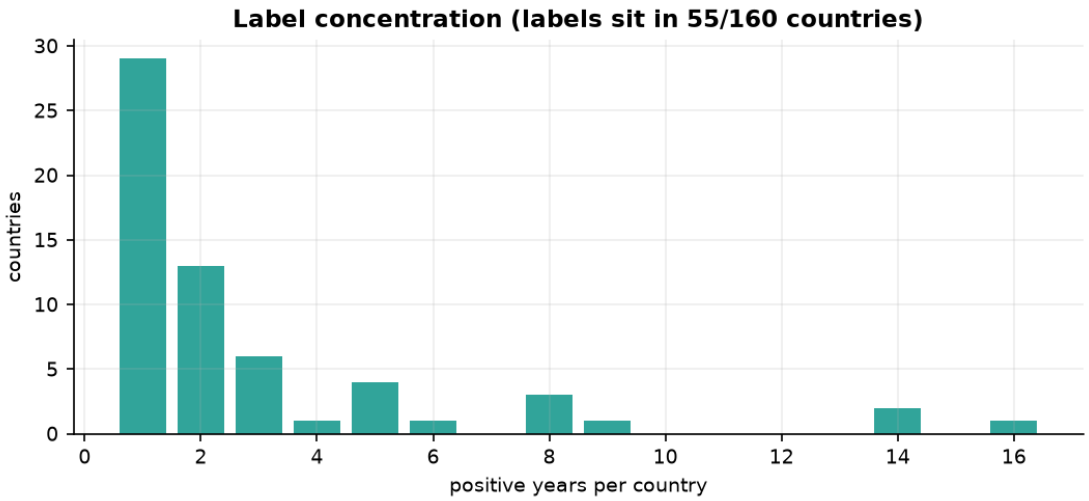


Figure 9: Distribution of positive years per country among countries with at least one positive. Authors' plot from the assembled modelling matrix.

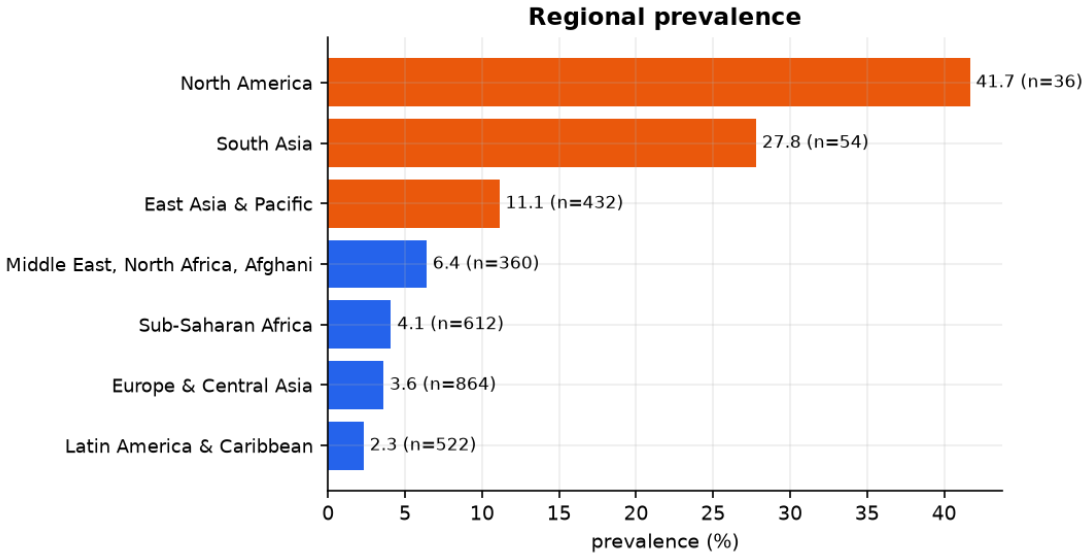


Figure 10: Prevalence by World Bank region. Authors' plot; region names follow the World Bank classification [32].

Table 16: Discrimination at the country-year level. Brackets give 95 per cent bootstrap intervals on the temporal holdout. Expected calibration error for the champion is 0.025 (Brier 0.0555).

Model	PR-AUC (time)	95% CI	ROC-AUC (time)	PR-AUC (country)
Logistic regression	0.461	[0.34, 0.58]	0.844	0.381
Decision tree	0.376	[0.28, 0.48]	0.871	0.243
Random forest	0.495	[0.38, 0.62]	0.874	0.322
Extra trees	0.494	[0.38, 0.60]	0.866	0.353
XGBoost	0.429	[0.32, 0.54]	0.846	0.268
LightGBM	0.471	[0.36, 0.58]	0.846	0.280

Because deployment means an unseen country rather than an unseen year, the champion is Logistic Regression. The inversion is the more informative result: with eight retained features the relationships are close enough to additive that a regularised linear model transfers best, and the flexible ensembles pay for their extra capacity in exactly the setting the project cares about. The single decision tree is weakest on both schemes, which confirms that the signal is not concentrated in one or two splits.

2. Discussion

All measured results above are for one task only: given features from year t, predict whether glanders is documented in that country in year $t+1$. Logistic Regression reaches a precision-recall AUC of 0.461 against a base rate of 5.87% on the forward split, and holds 0.381 across unseen countries, with calibration error 0.025. The gap between those two figures is smaller than the full-feature model showed, which is the practical gain from selecting features on conditional evidence. The signal it learns is surveillance rather than biology: the leading features are population, connectivity and reporting measures, the label concentrates in countries with reference laboratories, and the best-scoring cells are India, the United States and Australia.

Chapter 6

Conclusion and Future Work

1. Conclusion

This project set out to build a usable warning system for glanders: to establish whether population-level risk could be predicted from public data, and how a prediction reaches the person who owns the animals. Four sources new to the disease formed a country-year panel across 160 equine-holding countries, and six model families were trained under three validation schemes. Beside the model, the platform is split into three components with fixed contracts: a risk engine, a RAG question-answering layer and an alert service.

Prediction is possible, and the performance is moderate: Logistic Regression reaches a precision-recall AUC of 0.461 against a base rate of 5.87% on a forward split, about an 8-fold lift with a calibration error of 0.025, and holds 0.381 across unseen countries. The skill is fragile across geography, and the decline concentrates where early warning would help most.

What the model learns is less encouraging. Three analyses converge: the highest-ranked features are population size and reporting capacity rather than biological drivers, the label concentrates in countries operating reference laboratories, and the best-scoring cells have the strongest publishing record. It predicts where glanders will be documented, not where it is.

2. Limitations

Five limits cut across the platform; each is stated once and held for every claim that follows.

1. A positive means glanders was reported in the literature for that country-year; a negative may mean no disease or simply no report. With 169 positives at country-year resolution, leave-country-out skill is weaker than temporal skill, and the drop is largest where early warning would help most.

2. The 263 occurrence events come from an automated pipeline that has not been checked against a hand-annotated reference, so extractor error is not separately quantified.

3. For the RAG layer, retrieval quality, faithfulness, citation precision and refusal behaviour have fixed targets and an evaluation protocol, but no scored run against the guidance corpus or against real officer questions.

4. Thresholds, hysteresis and deduplication are designed against the calibration curve, yet SMS/WhatsApp providers, durable workers and a consented subscriber set are not yet in place, so delivery success and alert fatigue remain unmeasured.

5. A logged acceptance from a provider does not prove that an owner saw the warning or took action; acknowledgement and escalation are not yet part of the service.

3. Future Work

Five next steps cover the model, the RAG layer and the alert service.

1. Treat surveillance effort as a quantity, for example an offset on publication volume or a joint detection-and-occurrence model, so gains in score reflect disease rather than publishing, and cross-check the literature label against WOAH WAHIS where the two overlap [28].
2. Annotate a stratified sample of the 263 events for extractor precision and recall, then move toward district-month prediction for India once public district covariates can be assembled.
3. Score recall@5, faithfulness, citation precision and refusal rates on a fixed question set, then ask district officers to rate answer usefulness on real questions.
4. Put durable storage and workers behind the rules core, integrate approved SMS/WhatsApp for a consented pilot, and keep monthly and yearly states separate once a monthly prediction pipeline exists.
5. Measure alert precision, missed events, messages per recipient, delivery latency and whether owners acknowledge HIGH alerts; escalate unacknowledged warnings only under a veterinary-approved policy.

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